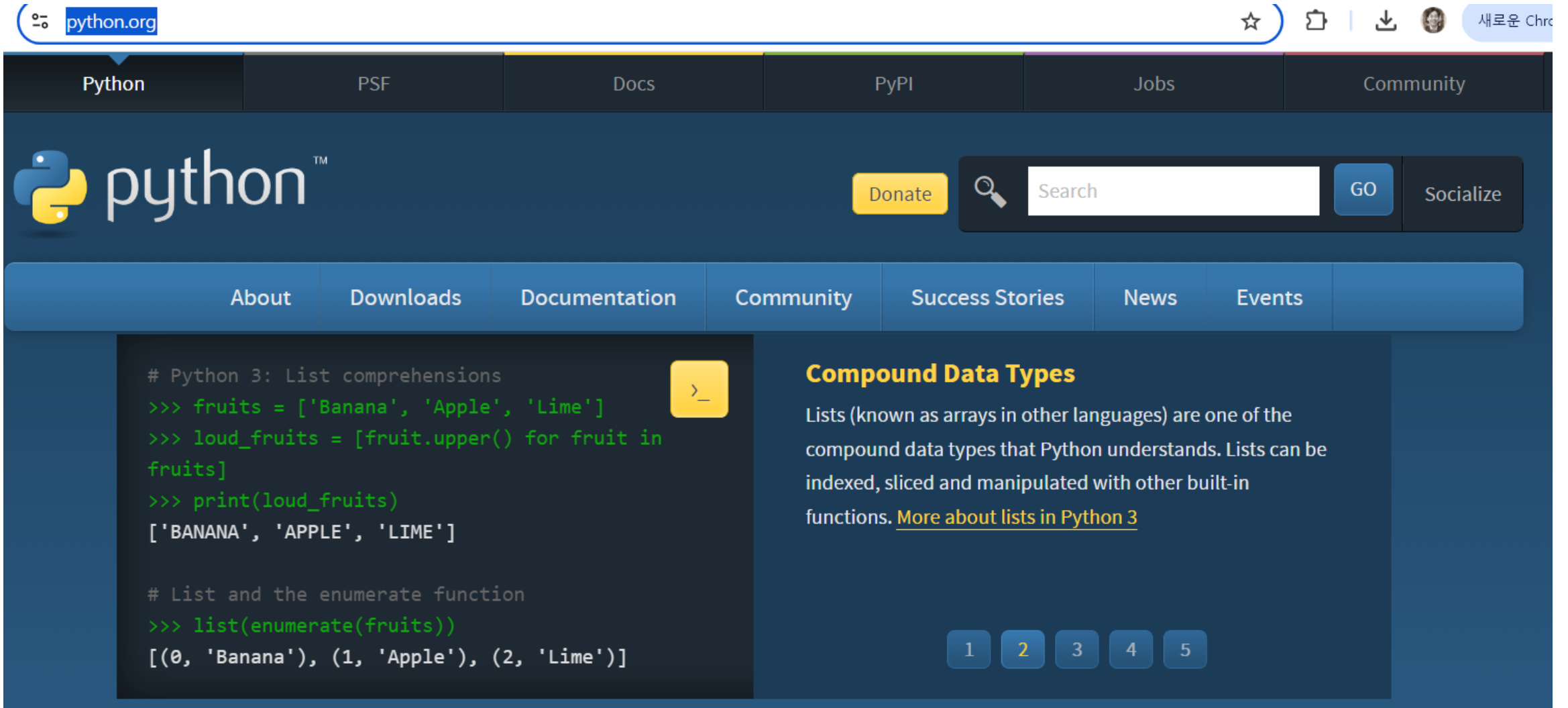


Python Basic

Installation

Python Installation



The image is a screenshot of the Python.org homepage. At the top, there is a browser address bar showing 'python.org'. Below the address bar is a dark navigation bar with links for 'Python', 'PSF', 'Docs', 'PyPI', 'Jobs', and 'Community'. The main header features the Python logo and the word 'python' in a large, white, sans-serif font. To the right of the logo is a yellow 'Donate' button, a search bar with a magnifying glass icon, a 'GO' button, and a 'Socialize' button. Below the header is a blue navigation bar with links for 'About', 'Downloads', 'Documentation', 'Community', 'Success Stories', 'News', and 'Events'. The main content area is divided into two columns. The left column contains a code editor with a dark background and a yellow 'Run' button. The code shows two Python snippets: one for list comprehensions and one for the enumerate function. The right column has a section titled 'Compound Data Types' in orange, followed by a paragraph explaining lists and a link to 'More about lists in Python 3'. At the bottom right, there is a row of five numbered buttons (1, 2, 3, 4, 5), with button 2 being highlighted.

python.org

Python PSF Docs PyPI Jobs Community

python™

Donate

Search GO Socialize

About Downloads Documentation Community Success Stories News Events

```
# Python 3: List comprehensions
>>> fruits = ['Banana', 'Apple', 'Lime']
>>> loud_fruits = [fruit.upper() for fruit in fruits]
>>> print(loud_fruits)
['BANANA', 'APPLE', 'LIME']

# List and the enumerate function
>>> list(enumerate(fruits))
[(0, 'Banana'), (1, 'Apple'), (2, 'Lime')]
```

Compound Data Types

Lists (known as arrays in other languages) are one of the compound data types that Python understands. Lists can be indexed, sliced and manipulated with other built-in functions. [More about lists in Python 3](#)

1 2 3 4 5

<https://www.python.org/>

VS Code Install

The image shows the Visual Studio Code website on the left and a screenshot of the VS Code editor on the right. The website has a navigation bar with links to Docs, Updates, Blog, API, Extensions, FAQ, and Learn. A 'Download' button is prominent. Below the navigation bar, a message states 'Version 1.92 is now available! Read about the new features and fixes from July.' The main content area features the text 'Code Editing. Redefined.' and a 'Download for Windows' button. A link for 'Web, Insiders edition, or other platforms' is also present. The editor window on the right shows a file explorer on the left with a project named 'MY-APP' containing various components. The main editor area displays a TypeScript file 'button.ts' with code for a button component. A modal dialog is open over the code, asking to 'Create a new button component' with 'Accept' and 'Discard' buttons. The bottom status bar shows 'main' and '0 ↓ 1 ↑'.

code.visualstudio.com

Visual Studio Code Docs Updates Blog API Extensions FAQ Learn

Search Docs

Download

Version 1.92 is now available! Read about the new features and fixes from July.

Free. Built on open source. Runs everywhere.

Code Editing. Redefined.

Download for Windows

[Web](#), [Insiders edition](#), or [other platforms](#)

By using VS Code, you agree to its [license](#) and [privacy statement](#).

EXPLORER

- MY-APP
 - components
 - actionbar
 - breadcrumbs
 - button
 - # button.css
 - TS button.ts
 - countBadge
 - dialog
 - dropdown
 - findinput
 - grid
 - hover
 - inputBox
 - .gitignore
 - .mailmap
 - .mention-bot
 - .yarnrc
 - yarn.lock

TS button.ts

```
1 interface ButtonProps {
2   onClick: () => void;
3   text: string;
4 }
5 const Button: React.FC<Props> = ({ onClick, text }) => {
6   return <button onClick={onClick}>{text}</button>;
7 };
8
9 export default Button;
```

Create a new button component

Accept Discard

Changed 9 lines

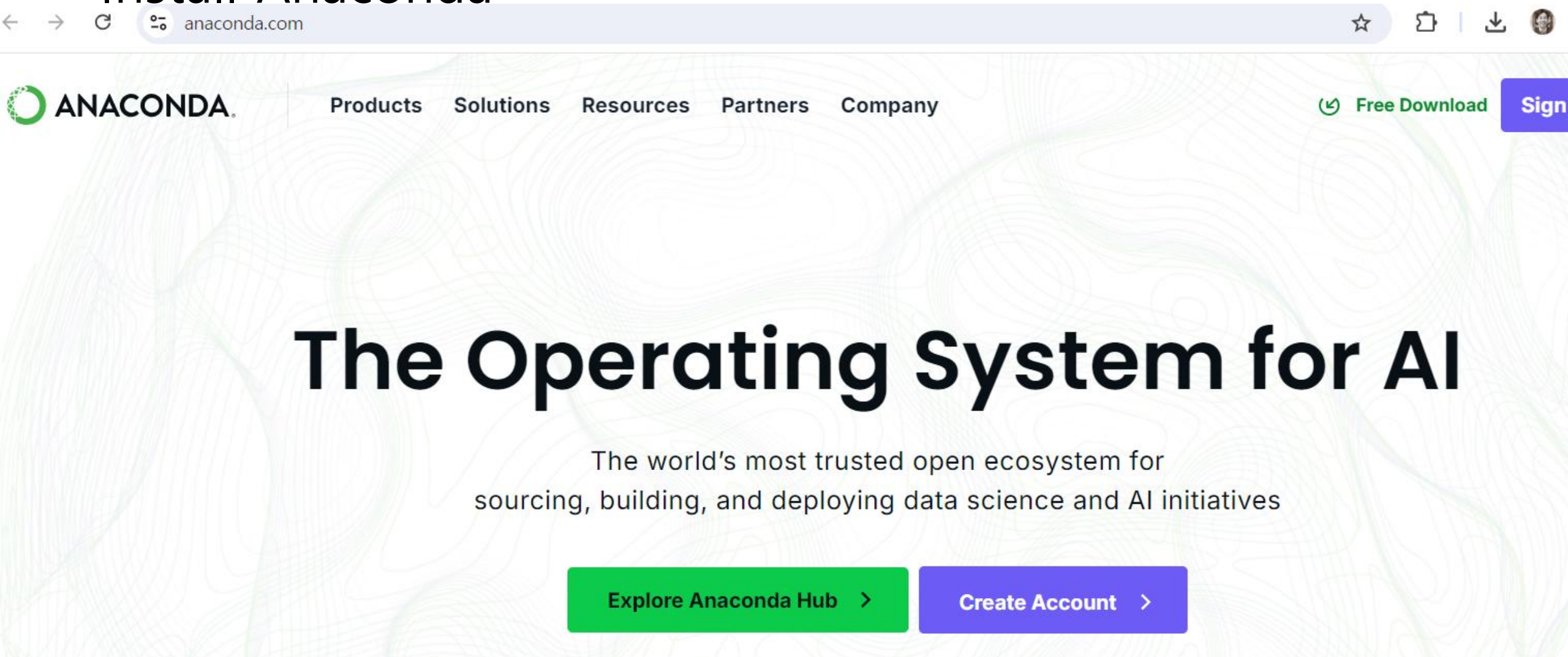
PROBLEMS OUTPUT TERMINAL

zsh

my-app git:(main)

<https://code.visualstudio.com/>

Install Anaconda

The image is a screenshot of the Anaconda website homepage. At the top, there is a browser address bar showing 'anaconda.com'. Below the address bar is the Anaconda logo on the left and a navigation menu with links for 'Products', 'Solutions', 'Resources', 'Partners', and 'Company' in the center. On the right side of the navigation bar, there is a green 'Free Download' button with a download icon and a blue 'Sign In' button. The main content area features a large, bold heading 'The Operating System for AI' centered on the page. Below this heading is a sub-headline: 'The world's most trusted open ecosystem for sourcing, building, and deploying data science and AI initiatives'. At the bottom of the main content area, there are two prominent buttons: a green 'Explore Anaconda Hub' button with a right-pointing chevron, and a blue 'Create Account' button also with a right-pointing chevron. The background of the page has a subtle, abstract pattern of green and white lines.

<https://www.anaconda.com/>

환경 설정

Open terminal

Anaconda power shell

Jupyter notebook

conda install jupyter notebook

Install numpy, pandas, matplotlib, seaborn, scikit-learn

conda install numpy

conda install pandas

conda install matplotlib

conda install seaborn

conda install scikit-learn

Basic Python Library

Computational Physics & Machine Learning 에서 필요한
기초적인 파이썬 문법 , 라이브러리

1. Python basic grammar – variable, data structure, loop, condition
2. Numpy -'numpy.ndarray'
3. Pandas s -DataFrame
4. Matplotlib – Data visualization
5. Scikit -learn – Data 분석에 필요한 다양한 데이터 셋, 알고리즘 제공

• Numpy

- Stands for Numerical Python
- 다차원 배열을 위한 기능 제공
- 선형 대수 연산과 같은 데이터를 다루는 데에 필수적인 수학적 계산이 가능
- Scikit -learn 에서 다루는 데이터의 기본 구조는 Numpy 배열 (numpy.ndarray)

```
In [12]: # np array and print
num = np.array(3)
print("num →", num)
print(type(num))

real = np.array(1.0)
print("real →", real)
print(type(real))

string = np.array('data')
print("string →", string)
print(type(string))

arr = np.array([1, 2, 3])
print("arr →", arr)
print(type(arr))
```

```
num → 3
<class 'numpy.ndarray'>
real → 1.0
<class 'numpy.ndarray'>
string → data
<class 'numpy.ndarray'>
arr → [1 2 3]
<class 'numpy.ndarray'>
```

• Pandas

- 데이터 처리와 분석을 위한 파이썬 라이브러리
- DataFrame 이라는 엑셀표와 비슷한 데이터 구조를 제공 (R의 data.frame 과 비슷하다)
- 대용량 데이터를 효율적으로 다루는 것이 가능
 - 엑셀 ; 데이터의 용량이 100MB 넘어가면 느려짐
 - Pandas; 최소 1GB, 많으면 100GB 가 넘는 데이터도 빠른 속도로 처리
- 파이썬의 문법을 그대로 가지고옴 , 직관적으로 함수가 구현되어 있어서 쉽게 읽을 수 있음

```
1 raw_data = pd.read_csv('prostate_dataset.txt', delimiter='#t')
```

```
1 raw_data.head(10)
```

	col	lcavol	lweight	age	lbph	svi	lcp	gleason	pgg45	lpsa	train
0	1	-0.579818	2.769459	50	-1.386294	0	-1.386294	6	0	-0.430783	T
1	2	-0.994252	3.319626	58	-1.386294	0	-1.386294	6	0	-0.162519	T
2	3	-0.510826	2.691243	74	-1.386294	0	-1.386294	7	20	-0.162519	T
3	4	-1.203973	3.282789	58	-1.386294	0	-1.386294	6	0	-0.162519	T
4	5	0.751416	3.432373	62	-1.386294	0	-1.386294	6	0	0.371564	T
5	6	-1.049822	3.228826	50	-1.386294	0	-1.386294	6	0	0.765468	T
6	7	0.737164	3.473518	64	0.615186	0	-1.386294	6	0	0.765468	F
7	8	0.693147	3.539509	58	1.536867	0	-1.386294	6	0	0.854415	T
8	9	-0.776529	3.539509	47	-1.386294	0	-1.386294	6	0	1.047319	F
9	10	0.223144	3.244544	63	-1.386294	0	-1.386294	6	0	1.047319	F

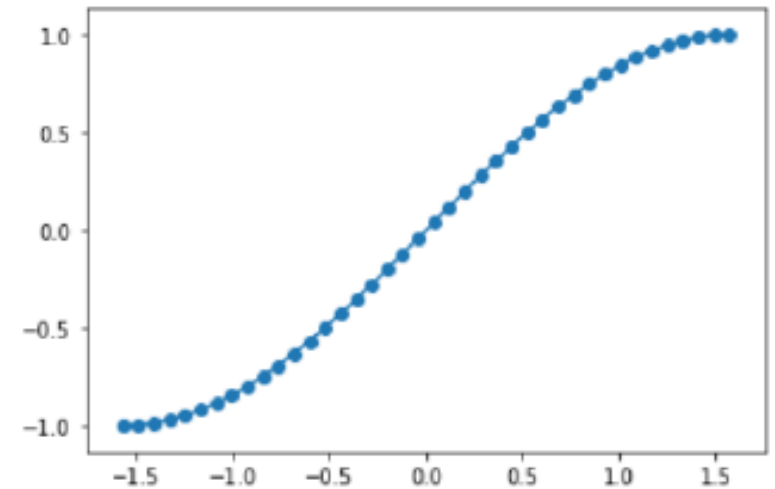
- Matplotlib

- 과학 계산용 그래프 라이브러리
- 선그래프, 히스토그램 등 지원

```
# plot 그리기  
  
y = np.sin(X)  
plt.plot(X, y)  
plt.scatter(X, y)  
plt.show()
```

- Seaborn

- Matplotlib 기반 통계 데이터 시각화 라이브러리
- 데이터 분포도, 히트맵, 박스플롯 그리기등



- Scikit-learn (sklearn)
 - 머신러닝 알고리즘 제공

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 import mglearn
5
6 from sklearn.model_selection import train_test_split
7 from sklearn.linear_model import LinearRegression
8 from sklearn.metrics import mean_squared_error
```

GitHub 계정 생성 & git 설치

The screenshot shows the GitHub homepage in a web browser. The browser's address bar displays 'github.com'. The page header includes the GitHub logo, 'Dashboard', a search bar with the placeholder 'Type to search', and navigation icons. On the left sidebar, there's a 'Top repositories' section with a 'New' button and a search bar. Below it, two repositories are listed: 'nicesw77/CPP' and 'nicesw77/Example-and-Practice'. The main content area is titled 'Home' and features a 'Start writing code' button. It contains two primary cards: 'Start a new repository for nicesw77' and 'Introduce yourself with a profile README'. The first card has a form for 'Repository name *' and options for 'Public' and 'Private' visibility. The second card shows a preview of a README file for 'nicesw77 / README.md' with a 'Create' button. On the right, there's a 'GitHub Models' banner encouraging users to join the 'Limited Public Beta' and a section titled 'Explore repositories' listing 'raysan5 / raylib' and 'facebook / react'.

Top repositories

Find a repository...

nicesw77/CPP

nicesw77/Example-and-Practice

GitHub

1. 내 소스를 저장
2. 소스코드 공유
3. 협업하는 공간

<https://github.com/>

GitHub 계정 생성 & git 설치: YouTube 코딩알려주는 누나
<https://www.youtube.com/watch?v=lelVripbt2M>



Git is a **free and open source** distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is **easy to learn** and has a **tiny footprint with lightning fast performance**. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like **cheap local branching**, convenient **staging areas**, and **multiple workflows**.



About

The advantages of Git compared to other source control systems.



Documentation

Command reference pages, Pro Git book content, videos and other material.



Downloads

GUI clients and binary releases for all major platforms.



Community

Get involved! Bug reporting, mailing list, chat, development and more.



git을 통해 source code upload

Q & A